Jacob Platin

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EDUCATION AND SKILLS

University of Pennsylvania, School of Engineering and Applied Science — **Philadelphia, PA**
*MSE in **Robotics** (Machine Learning Specialty), GPA 3.7/4.0 (Magna Cum Laude)* — Aug 2017 – May 2022
- **Relevant Coursework**: Machine Learning, Network System Design, Computer Vision, Deep Learning

University of Pennsylvania, School of Engineering and Applied Science — **Philadelphia, PA**
BSE, Majors in ***Computer Science*** & ***Economics***, *GPA 3.5/4.0 (Cum Laude)* — Aug 2017 - May 2022
- **Relevant Coursework**: Cloud Computing/Scalability, Econometrics, Game Theory, Data Structures, Software Design

ETH Zurich, Departments of Computer Science and Economics — **Zurich, Switzerland**
Exchange Program, GPA 3.75/4.0 — Sep 2019 - Dec 2019
- **Relevant Coursework**: Computer Architecture, Reliable Artificial Intelligence, Wireless/Mobile Computing

EXPERIENCE

Google | *Senior Software Engineer (Machine Learning)* | Kirkland, WA — Mar 2026 – Present
- Leading bring-up and inference optimization efforts for Qwen3.5, DeepSeek, and Kimi-K2.6, including theoretical roofline and headroom analysis in both PyTorch and JAX
- Exploring novel quantization techniques to boost performance while preserving accuracy at low precision, including creating and modifying existing attention, general matmul and GMM Pallas kernels

Google | *Software Engineer III (Machine Learning)* | Kirkland, WA — Feb 2025 – Mar 2026
- Led weight + activation and KV cache quantization efforts on Google's TPU and GPU OSS vLLM stack to increase end-to-end throughput by 90% while preserving 99% quality
- Collaborated internally and externally to implement new models, including DeepSeekV3 and Llama4, and new features, including multi-latent attention and mixture-of-experts, into the stack

Microsoft | *Software Engineer II (Machine Learning)* | Redmond, WA — Dec 2023 – Feb 2025
- Integrated internal speech models and LLMs to achieve SOTA multi-modal model performance
- Led two sub-teams dedicated to increasing model size and training speed at minimum compute cost
- Fostered an inclusive and growth-oriented team by organizing paper readouts, relaxation retreats, and learning sessions

Microsoft | *Software Engineer (Machine Learning)* | Redmond, WA — Aug 2022 – Nov 2023
- Led efforts on optimizing model size, performance, and throughput for Microsoft's latest speech recognition models by applying state-of-the-art sharding, networking, and architecture-based techniques
- Owned and enhanced the software framework that 200 members of the Azure AI Speech team used to train models

Unity Technologies | *Software Engineer Intern (Robotics)* | Seattle, WA — May 2021 - Aug 2021
- Integrated an inverse kinematics solver directly into the Unity engine using linear algebra and robotics principles
- Collaborated with NVIDIA to implement realistic, physics-based joint controllers for robotic simulations (in VR)

NVIDIA | *Software Engineer Intern* | Redmond, WA — Feb 2021 - May 2021
- Developed and shipped a cloud-based searching solution for game meta-data using Elasticsearch, GraphQL and AWS that improved query latency by 35% and reduced the existing codebase size by 20%

PUBLICATIONS

Microsoft | *Phi-4-Mini Technical Report* | arxiv.org/abs/2503.01743 — Mar 2025
- Integrated Microsoft's SOTA multi-modal language model with vLLM to optimize inference performance

LEADERSHIP & MENTORSHIP

Rainier Athletes | *Mentor* — Oct 2025 – Present
- Mentoring a middle school student to encourage his future success in scholastics, personal growth, and sports

TAMID Group | *Mentor* — Dec 2024 – Present
- Coach current TAMID students on how best to navigate their undergraduate experience by utilizing personal experience
- Guide multiple students through career exploration while helping them develop necessary skills to prepare for SWE/ML

Global Mentoring Initiative | *Mentor* — Aug 2023 – Present
- Provide career and technical mentorship to international students from disadvantaged backgrounds